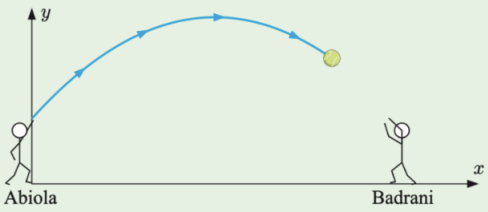
Chapter 2 : Quadratics

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I - Quadratic equations

Investigation

Abiola and Badrani are standing 40 metres apart, throwing a ball between them. When Abiola throws the ball, it travels in a smooth arc. At the time when the ball has travelled x metres horizontally towards Badrani, its height is y metres.







x (m)	0	5	10	15	20	25	30
y (m)	1.25	10	16.25	20	21.25	20	16.25

Things to think about:

a Use technology to plot these points.

b What *shape* is the graph of y against x?

c What is the maximum height reached by the ball?

d What *formula* gives the height of the ball when it has travelled x metres horizontally towards Badrani?

e Will the ball reach Badrani before it bounces?

Definition

A quadratic equation is an equation of the form

$$ax^2 + bx + c = 0 \quad \text{where } a, b, c \text{ are constants, } a \neq 0.$$

Definition

A quadratic function is a function of the form

$$y = ax^2 + bx + c, \quad a \neq 0.$$

Remark. Quadratic functions are members of the family of **polynomials**. The first few members of this family are shown in the table :

Polynomial function	Type
$y = ax + b,\ a \neq 0$	linear
$y = ax^2 + bx + c,\ a \neq 0$	quadratic
$y = ax^3 + bx^2 + cx + d,\ a \neq 0$	cubic
$y = ax^4 + bx^3 + cx^2 + dx + e,\ a \neq 0$	quartic

To solve quadratic equations we have the following methods to choose from :

— factorise the quadratic and use the **Null Factor law :**

$$\text{If } ab = 0 \text{ then } a = 0 \text{ or } b = 0.$$

— complete the square

— use the quadratic formula

— use technology

Definition

The roots or solutions of

$$ax^2 + bx + c = 0$$

are the values of x which satisfy the equation.

Example

Is 2 a root of $x^2 - 3x + 2$?

Example

Solve for x :

a) $3x^2 + 5x = 0$

b) $x^2 = 5x + 6$

Example

Solve for x :

a) $4x^2 + 1 = 4x$

b) $6x^2 = 11x + 10$

Example

Solve for x :

a) $(x+2)^2 = 7$

b) $(x-1)^2 = -5$

Example

Solve for exact values of x by completing the square : $x^2 + 4x + 1 = 0$

$x^2 + 4x + 1 = 0$

II - Quadratic formula

Property

If $ax^2 + bx + c = 0$ and $\Delta = b^2 - 4ac$

— If $\Delta = 0$, $x = \dfrac{-b}{2a}$ is **the only solution** (a repeated or double root).

— If $\Delta > 0$, $\sqrt{\Delta}$ is a positive real number, so there are **two distinct real roots** :

$$x = \frac{-b + \sqrt{\Delta}}{2a} \quad \text{and} \quad x = \frac{-b - \sqrt{\Delta}}{2a}$$

— If $\Delta < 0$, $\sqrt{\Delta}$ is not a real number and so there are **no real roots**.

— If a, b, c are rational and Δ is a **square** then the equation has two rational roots which can be found by factorisation.

Proof

$$ax^2 + bx + c = 0, \quad a \neq 0$$

$$\Rightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \{\text{dividing each term by } a\}$$

$$\Rightarrow x^2 + \frac{b}{a}x = -\frac{c}{a}$$

$$\Rightarrow x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \quad \{\text{completing the square}\}$$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad \{\text{factorising}\}$$

$$\Rightarrow x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

Solve for x :

a) $x^2 - 2x - 6 = 0$

b) $2x^2 + 3x - 6 = 0$

Example

Consider the equation $kx^2 + (k+3)x - 1 = 0$. Find the discriminant Δ and draw its sign diagram. Hence, find the value of k for which the equation has :

[label=)]

1. two distinct real roots
2. two real roots
3. a repeated root
4. no real roots

III - Quadratic functions

Definition

A quadratic function has the form

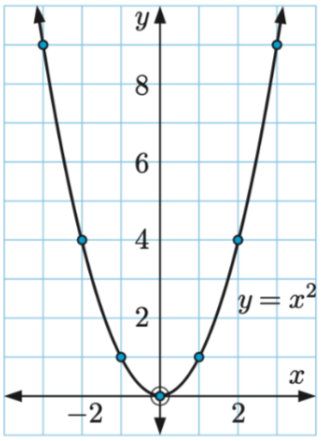
$$y = ax^2 + bx + c \quad \text{where } a \neq 0.$$

The simplest quadratic function is

$$y = x^2$$

Its graph can be drawn from a table of values :

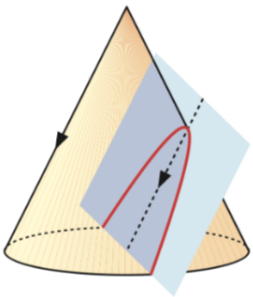
x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9



The graph of a quadratic function is called a **parabola**.

The parabola is one of the **conic sections**, the others being circles, hyperbolae and ellipses.

They are called conic sections because they can be obtained by cutting a cone with a plane.



A parabola is produced by cutting the cone with a plane parallel to its slant side.

Investigation

1. (a) Use technology to help you sketch :

$$y = (x-1)(x-3), \quad y = 2(x-1)(x-3), \quad y = -(x-1)(x-3),$$

$$y = -3(x-1)(x-3), \quad y = -\tfrac{1}{2}(x-1)(x-3)$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of a in $y = a(x-1)(x-3)$?

2. (a) Use technology to help you sketch :

$$y = 2(x-1)(x-4), \quad y = 2(x-3)(x-5), \quad y = 2(x+1)(x-2),$$

$$y = 2x(x+5), \quad y = 2(x+2)(x+4)$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of p and q in $y = 2(x-p)(x-q)$?

3. (a) Use technology to help you sketch :

$$y = 2(x-1)^2, \quad y = 2(x-3)^2, \quad y = 2(x+2)^2, \quad y = 2x^2$$

   (b) Find the x-intercepts for each function in (a).

   (c) What is the geometrical significance of p in $y = 2(x-p)^2$?

4. Copy and complete :

   — If a quadratic has the form $y = a(x-p)(x-q)$ then it ... the x-axis at ...

   — If a quadratic has the form $y = a(x-p)^2$ then it ... the x-axis at ...